KGiSL MicroCollege
Genius as a Service (GaaS)
TECHNICAL TEAM

KGCAS
co-Kreate your Genius

KG COLLEGE OF ARTS AND SCIENCE
Autonomous Institution | Affiliated to Bharathiar University
Accredited with A++ Grade by NAAC
ISO 9001:2015 Certified Institution
KGiSL Campus, Saravanampatti, Coimbatore – 641 035

C PROJECT REPORT ON

Matrix operation tool

SUBMITTED TO

The Technical Department

KG College of Arts and Science

SUBMITTED BY

Student Name	RollNo.	Course & Batch
Harini	25BCS116	
Hindhu priya	25BCS117	
Iniyha	25BCS118	
Jaiakash	25BCS119	
Janani sree	25BCS120	

Academic Year: **2025 – 2026**

Semester: **I**

DECLARATION

I hereby declare that the project titled “Matrix Operation Tool” is an original work carrie by me as part of my academic requirements. This project has been developed to perform matrix operations such as matrix addition, subtraction, multiplication, and transpose using programming concepts.

Student Name	RollNo.	Course & Batch	Signature
Harini	25bcs116		Harini
Hindhu priya	25bcs117		Hindhu Priya
Iniyha	25bcs118		Iniyha d.k
Jaiakash	25bcs119		Jaiakash
Janani sree	25bcs120		Janani sree

Place: COIMBATORE

Date:_08/11/2025_

CERTIFICATE

This is to certify that the project titled "Matrix Operation Tool" is a bonafide work carried out of KG College as part of the academic requirements. The project has been completed under the guidance and supervision of the concerned faculty during the course of study.

This project deals with implementing various matrix operations such as matrix addition, subtraction, multiplication, and transpose using programming concepts. To the best of my knowledge, this work is original and has not been submitted earlier to any other institution or university for the award of any degree or certificate.

S. No.	Name	**Roll Number**
1.	Harini	25BCS116
2.	Hindhu priya	25BCS117
3.	Iniyha	25BCS118
4.	Jaiakash	25BCS119
5.	jananisree	25BCS120

Guide:
(Technical Trainer, KGM)

Headof theDepartment
(Technical Head, KGM)

Place: Coimbatore

Date:08/11/2025

ABSTRACT

Matrix Operation Tool is a simple program developed to perform basic mathematical operations on matrices using programming concepts. Matrices are widely used in mathematics, computer science, and engineering for solving complex problems and representing data in a structured form.

This project provides a tool that allows users to perform different matrix operations such as addition, subtraction, multiplication, and transpose. The program takes matrix values as input from the user, processes the required operation, and displays the result in an organized format. It helps users understand how matrix calculations can be implemented using programming logic and algorithms.

The main objective of this project is to demonstrate the practical implementation of matrix operations and improve problem-solving skills in programming. This tool can be useful for students and beginners to learn matrix concepts and their applications in a simple and effective way.In addition, the Matrix Operation Tool is designed with a user-friendly approach so that users can easily enter matrix values and select the required operation. The program ensures accurate calculations and displays the output clearly, helping users verify the results of matrix operations. This project also helps in understanding the importance of programming structures such as loops, arrays, and conditional statements while performing mathematical computations.

TABLE OF CONTENTS

S. No.	Topics	Page. No.
1	**Introduction**	**1**
1.1	Objective	1
1.2	Overview	1
1.3	Features	1
2	**Program Analysis**	2
2.1	Problem Definition	2
2.2	Project Overview	2
2.3	Scopes and Limitations	2
3	**System Design**	**2**
3.1	Modules	3
4	**Methodology**	**3**
4.1	Problem Analysis	3
4.2	System Design	3
4.3	Program Development	3
4.4	Testing and Validation	3
4.5	Result and Documentation	4
4.6	Algorithm	4
4.7	Flowchart	4-5
5	**Source Code**	**6-8**
6	**Input and Output Screens**	**9**
7	**Discussion and Conclusion**	**10**
7.1	Discussion	10
7.2	Conclusion	10
8	**References**	**10**

1. INTRODUCTION

1.1 Objective

- In addition, the Matrix Operation Tool is designed with a user-friendly approach so that users can easily enter matrix values and select the required operation. The program ensures accurate calculations and displays the output clearly, helping users verify the results of matrix operations. This project also helps in understanding the importance of programming structures such as loops, arrays, and conditional statements while performing mathematical computations.

1.2 Overview

This system allows users to:

- The Matrix Operation Tool is a programming-based application designed to perform basic operations on matrices in a simple and efficient way. Matrices play an important role in mathematics, engineering, and computer science for representing and processing numerical data.

1.3 Features

- Performs basic matrix operations such as
- addition, subtraction, multiplication, and
- transpose.
- Allows the user to enter matrix elements
- easily through input.

  Displays the result of the matrix operation in

  a clear and organized format.

  Uses programming concepts like arrays,

  loops, and conditional statements.

2.PROGRAM ANALYSIS

2.1 Problem Definition

Matrix calculations are widely used in mathematics, computer science, and engineering, but performing these operations manually can be time-consuming and prone to errors, especially when dealing with large matrices. Many students also find it difficult to understand how matrix operations work in programming

2.2 Project Overview

The matrix operation tool simplifies matrix calculation by allowers users to perform basic operation quickly and accurately using programming The Matrix Operation Tool is a program that performs basic matrix operations like addition, subtraction, multiplication, and transpose. It allows users to enter matrix elements and quickly get accurate results, making calculations easier and faster while reducing errors.

2.3 Scopes and Limitations

Scope:

- Helps perform basic matrix operations quickly.

Limitations:

- The program is limited to basic matrix
- operations and does not support advanced matrix calculations.

  It can only handle matrices within the size limits defined in the program.

3.SYSTEM DESIGN

3.1 Modules

The Attendance Management System is divided into multiple functional modules, each handling a specific part of the program. These modules work together to simplify the attendance process and make the system more efficient and user-friendly.

Module Name	Description
input module	To accept matrix elements from the user.
output module	display the results of matrix operations.

4. METHODOLOGY

The methodology of the Matrix Operation Tool involves taking input matrices from the user, processing them according to the selected operation, and displaying the result. The program uses step-by-step computational logic to perform addition, subtraction, multiplication, or transpose. Each operation is executed using loops and conditional statements to ensure accurate and efficient calculation. This structured approach makes the tool user-friendly and reliable for solving matrix problems.

Development, Testing & Validation, and Result & Documentation.

4.1. Problem Analysis

Matrix Operation Tool is designed to simplify and automate the process of performing matrix operations. Traditionally, matrix calculations like addition, subtraction, multiplication, and transpose require careful manual computation, which can be time-consuming and prone to errors. By using this tool, users can input matrices of various sizes and instantly obtain accurate results.

4.2. System Design

Designing thestructureof files and functions used in the program.

4.3. Program Development

Developed inClanguage using functions and file-handling concepts for data storage.

4.4. Testing and Validation

Testedwith sampledata to ensure correct input/output operations and accuracy.

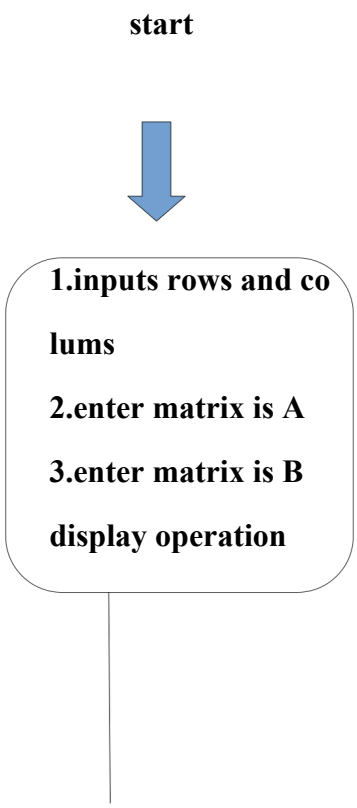
4.5. Result and Documentation

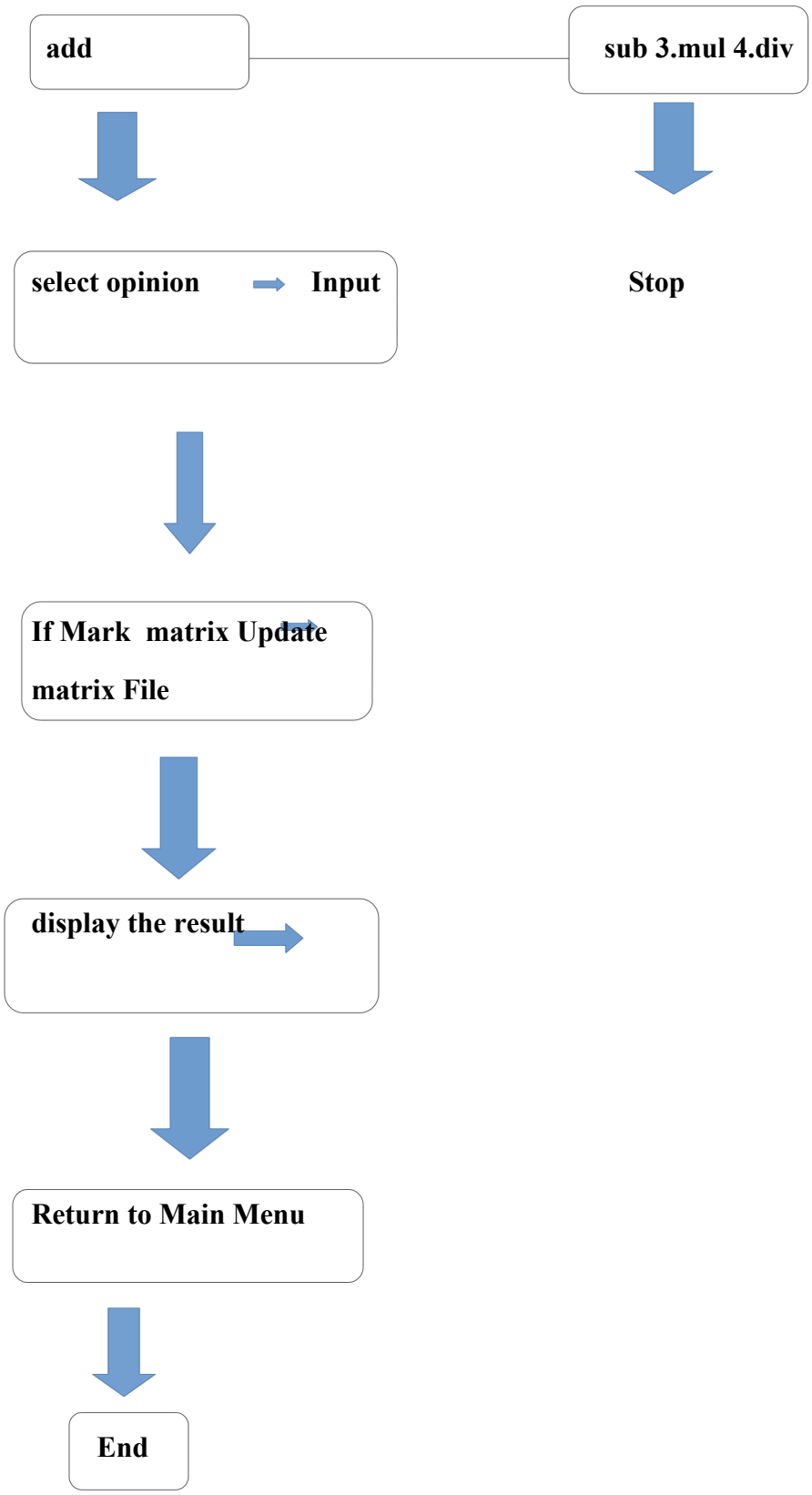
The tool accurately performs matrix operations, and the documentation explains its purpose, modules, algorithm, and results clearly.

Algorithm:
1.start
2.Enter matrix size and elements
3.Select the required matrix operation
4.Perform the operation
5.Display the result
6.Stop

Flow chart:

Matrix Operation Tool is a program used to perform basic matrix calculations such as addition, subtraction, multiplication, and transpose using programming.




add
sub 3.mul 4.div
select opinion
Input
Stop
If Mark matrix Update
matrix File
display the result
Return to Main Menu
End

5.SOURCE CODE

```
#include <stdio.h>

int main() {
        int  A[10][10],
B[10][10],     R[10]
[10], r, c, i, j, k, ch;
    printf("Rows and
columns:            ");
scanf("%d%d",&r,
&c);
        printf("Matrix
A:\n");
for(i=0;i<r;i++)
for(j=0;j<c;j++)
scanf("%d",&A[i]
[j]);
        printf("Matrix
B:\n");
for(i=0;i<r;i++)
for(j=0;j<c;j++)
scanf("%d",&B[i]
[j]);
         printf("1.Add
2.Sub            3.Mul
4.Transpose
A\nChoice:         ");
scanf("%d",&ch);

             if(ch==1){
for(i=0;i<r;i++){
for(j=0;j<c;j++){
R[i][j]=A[i][j]+B[i]
[j];         printf("%d
",R[i][j]);             }
printf("\n"); } }
      else  if(ch==2){
for(i=0;i<r;i++){
for(j=0;j<c;j++){
R[i][j]=A[i][j]-B[i]
[j];         printf("%d
",R[i][j]);             }
printf("\n"); } }
      else  if(ch==3){
for(i=0;i<r;i++){
for(j=0;j<c;j++){
R[i][j]=0;
for(k=0;k<c;k++)
```

```
Rresult[i][j] += A[i][k] * B[k][j];
            }
        printf("Result of Multiplication:\n");
        for(i=0;i<r1;i++){
            for(j=0;j<c2;j++)
                printf("%d ", result[i][j]);
            printf("\n");
        }
        break;

    case 4: // Transpose of A
        printf("Transpose of Matrix A:\n");
        for(i=0;i<c1;i++){
            for(j=0;j<r1;j++)
                printf("%d ", A[j][i]);
            printf("\n");
        }
        break;

    default:
        printf("Invalid choice!\n");
  }

  return 0;
}
```

```
Matrix A:
1  2
3  4
Matrix B:
5  6
7  8

Addition (Choice: 1)

Result of Addition:
   6   8
  10  12

Subtraction (Choice: 2)

Result of Subtraction:
  -4  -4
  -4  -4

Multiplication (Choice: 3)

Result of Multiplication:
  19  22
  43  50

Transpose of Matrix A (Choice: 4)

Transpose of Matrix A:
   1   3
   2   4
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The Matrix Operation Tool helps perform basic matrix calculations using programming. It makes matrix operations faster, easier, and more accurate.

7.2 Conclusion

The matrix Operation Tool project successfully demonstrates how basic matrix operations such as addition, subtraction, multiplication, and transpose can be implemented using programming concepts. This project helps in understanding the use of arrays, loops, and conditional statements while performing matrix calculations. Overall, the tool provides a simple and effective way to perform matrix operations and improves both programming knowledge and mathematical understanding.

8. REFERENCES

- Balagurusamy, E. *Programming inANSIC*
- Yashavant Kanetkar, Let Us C Programming,
- E. Balagurusamy, Programming in ANSI C,

  McGraw Hill Education.,